

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background.

A COMPREHENSIVE COVERAGE

MODULE - 16

AIR POLLUTION - 3

GLOBAL DIMMING

- Global Dimming is the reduction in the amount of solar radiation reaching the Earth's surface that has been observed since the 1950s. The effect varies by location.
- Global Dimming is caused by an increase in particulates and aerosols (sulfate aerosols) in the atmosphere due to human action.
- Global Dimming has interfered with the water cycle by reducing evaporation and led reduced rainfall in some areas
- Aerosols and particulates absorb solar energy and reflect sunlight back into space.



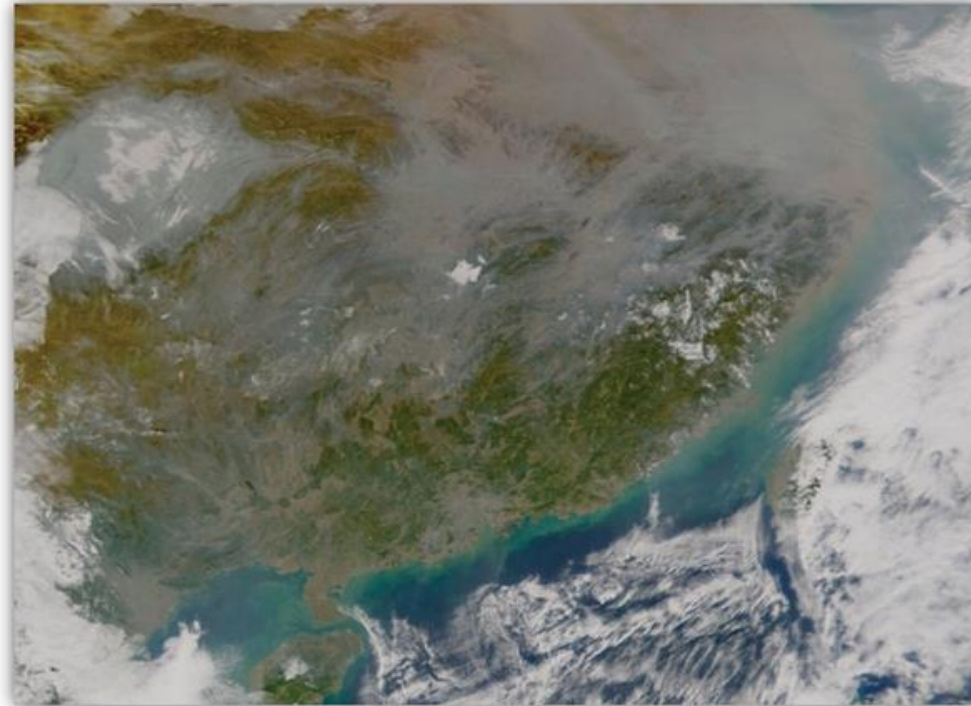
GLOBAL DIMMING

- The particles can also become nuclei for cloud droplets.
- Increased pollution causes more particulates and thereby creates clouds consisting of a greater number of smaller droplets (that is, the same amount of water is spread over more droplets).
- The smaller droplets make clouds more reflective, so that more incoming sunlight is reflected back into space and less reaches the Earth's surface.
- Clouds intercept both heat from the sun and heat radiated from the Earth. Usually during the daytime, the interception of sunlight predominates, giving a cooling effect.
- However, at night the re-radiation of heat to the Earth slows the Earth's heat loss and causes warming.



ASIAN BROWN CLOUD

- The Indian Ocean brown cloud or Asian brown cloud is a layer of air pollution that the northern Indian Ocean, India, and Pakistan.
- It appears every year between January and March.
- The cloud forms in the winter during which there is no rain to wash pollutants from the air.



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